

DEVELOPMENT OF PALATE

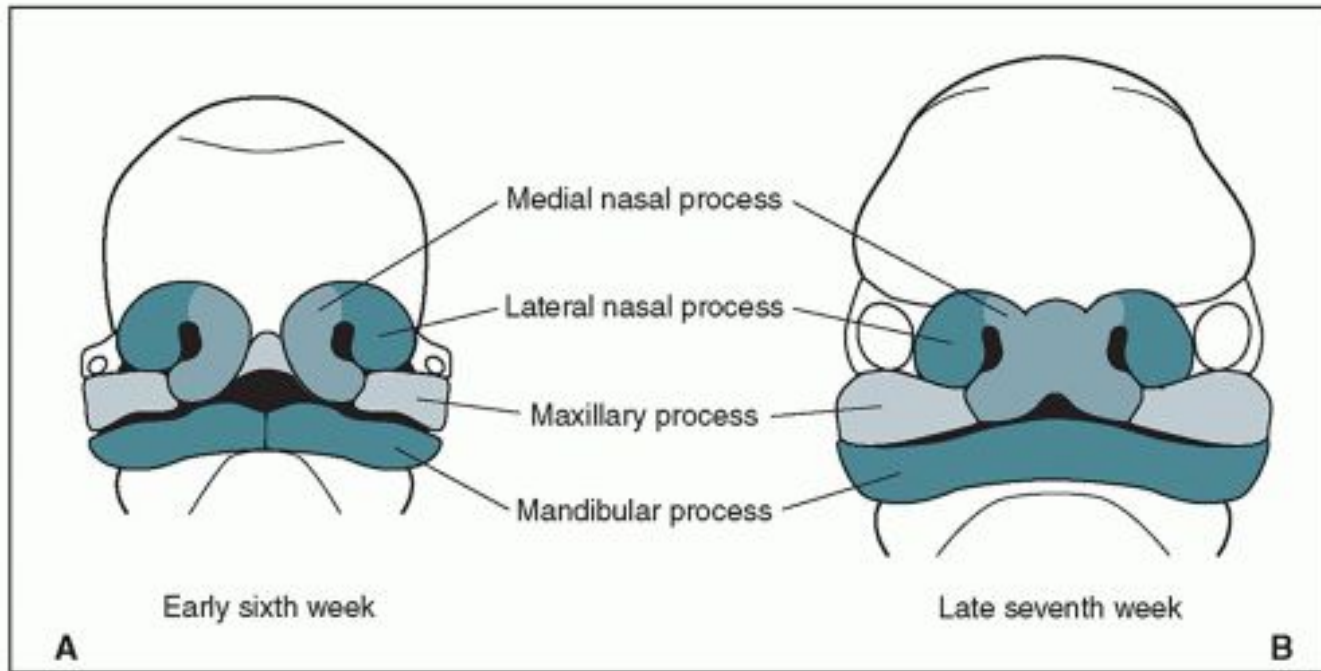
*DEPARTMENT OF ORAL
BIOLOGY DIKIOHS*

OBJECTIVES

- ☐ History
- ☐ Palatogenesis
- ☐ Development in embryo
- ☐ Clinical consideration
- ☐ Etiology
- ☐ Treatment

HISTORY

During the 5th week of embryo development, the first arch divides into two mandibular and two maxillary processes.



MAXILLARY PROCESS

- Maxilla
- Lateral upper lip
- ***Secondary palate***

MEDIAL PROCESS

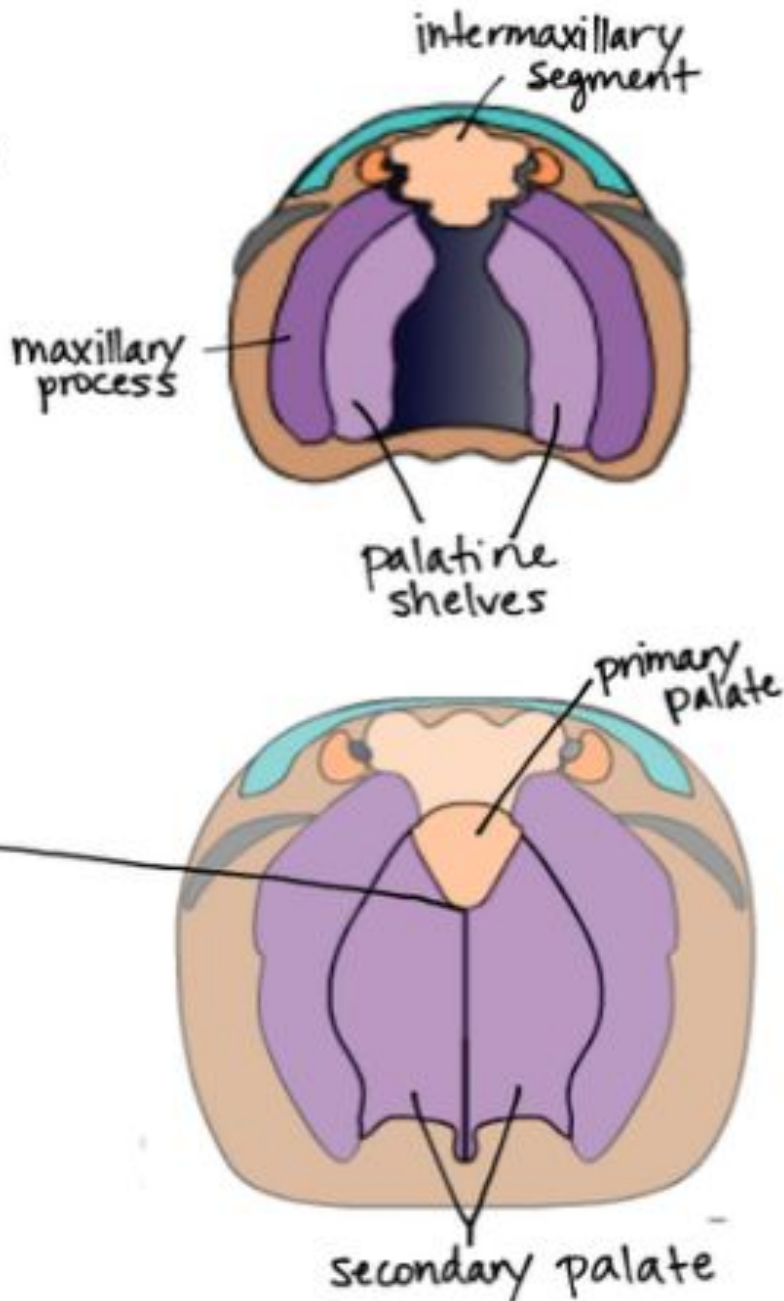
- Nose septum
- Philtrum
- ***Primary palate***

MANDIBULAR PROCESS

- Lower jaw
- Lower lip
- Chin

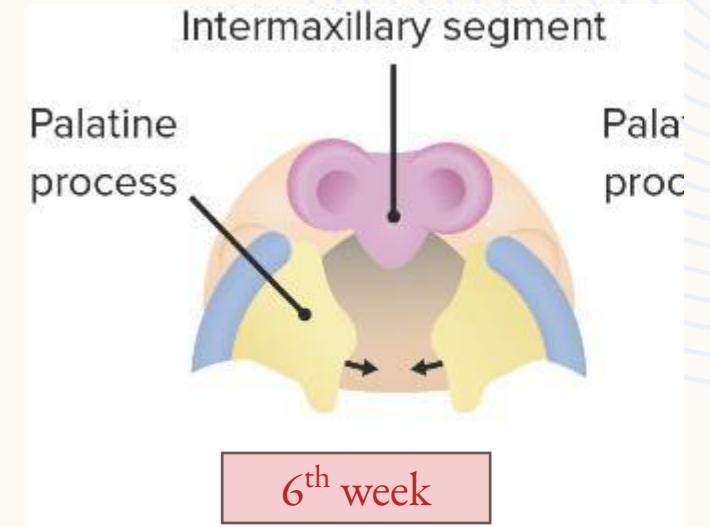
PALATOGENESIS

- PALATOGENESIS begins towards the end of 5th week and is completed about 12th week
- Palate developed from two primordia
 - **PRIMARY PALATE** : The primary palate or pre-maxilla includes that portion of the alveolar ridge containing the four incisors.
 - **SECONDARY PALATE** : The secondary palate includes the hard and soft palate. The hard palate is formed by the palatine processes of the maxillae and by the horizontal lamina of the palatine bones.



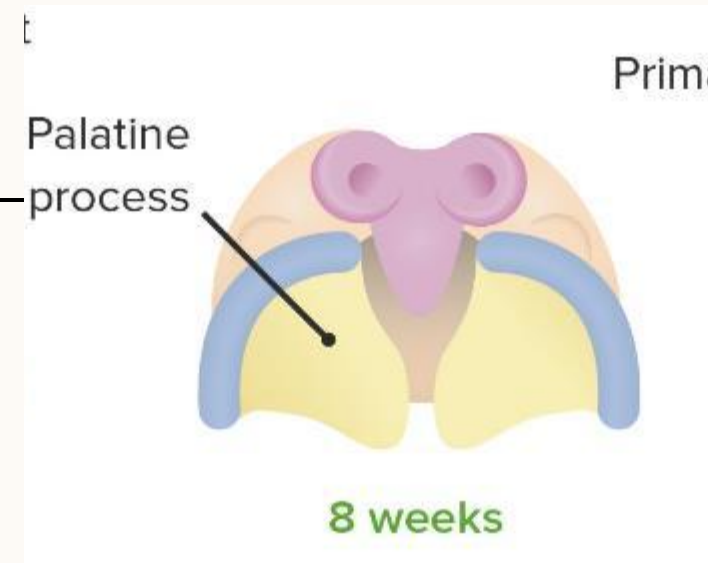
6th

By the 6th week the primary palate, formed by the two maxillary and two medial nasal processes give rise to intermaxillary segment, separates the developing oral and nasal cavities



7th

By the end of the 7th week the intermaxillary prominences grow into the oral cavity to form the primary palate and the philtrum, concluding upper lip and primary palate formation



8th

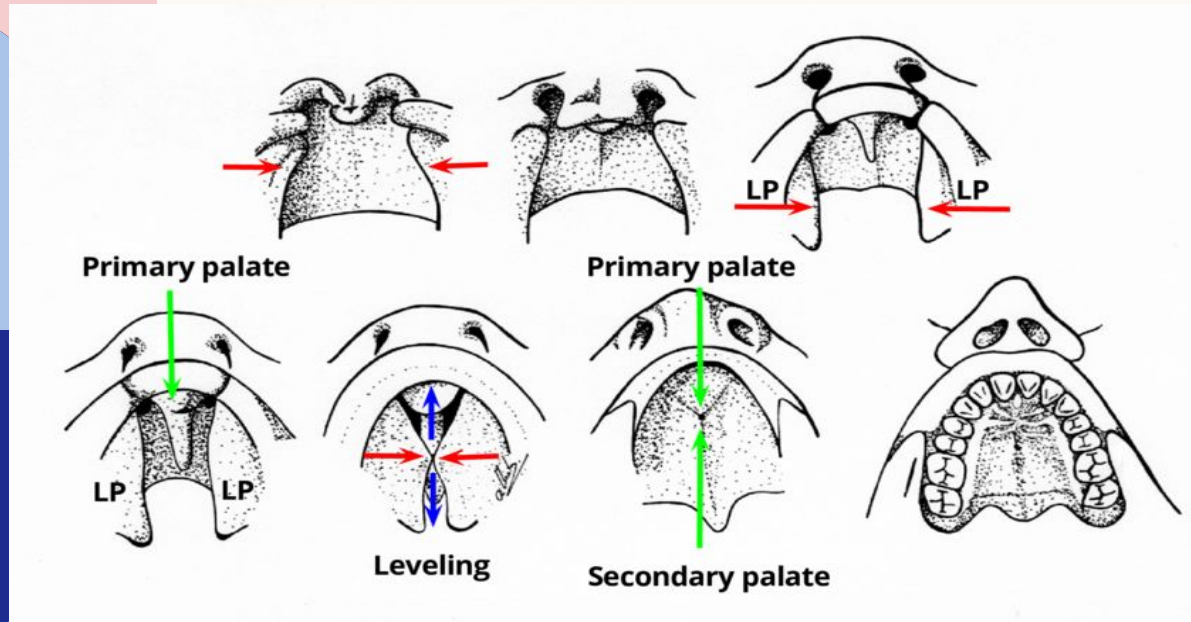
During 8th week the secondary palate forms as the palatine process of maxilla grows in upward and horizontal position the rest of the hard palate and all of the soft palate developed.
Fusion of palatine shelves elevates and fuse with each other above the tongue with the primary palate

ORIGIN OF PRIMARY PALATE

the medial nasal prominences - they merge forming the INTERMAXILLARY SEGMENT (medial nasal swelling)

ORIGIN OF SECONDARY PALATE

- maxillary prominences - 1st arch
the two shelves grow downwards vertically from the MxPs
- they eventually become horizontal and fuse down the midline when the tongue has descended (as head and jaws enlarge) and is no longer in the way

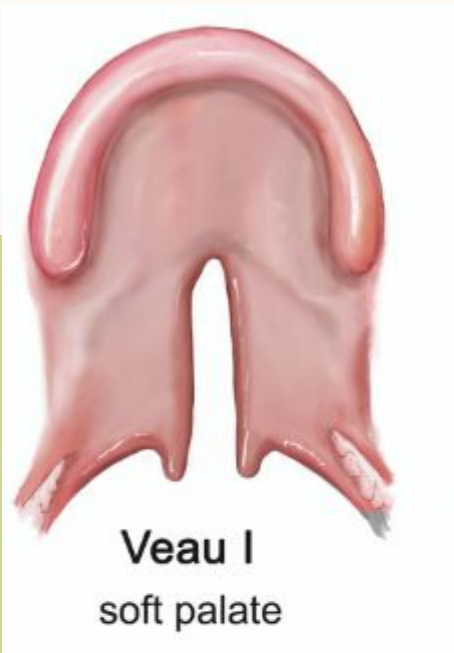


CLINICALS

CLEFT PALATE

VEAU CLASS I

Incomplete cleft only
involving soft palate



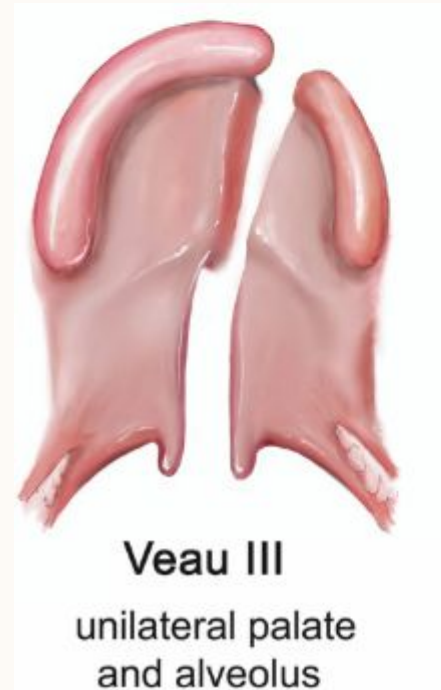
VEAU CLASS II

Cleft involving the hard
and soft palate



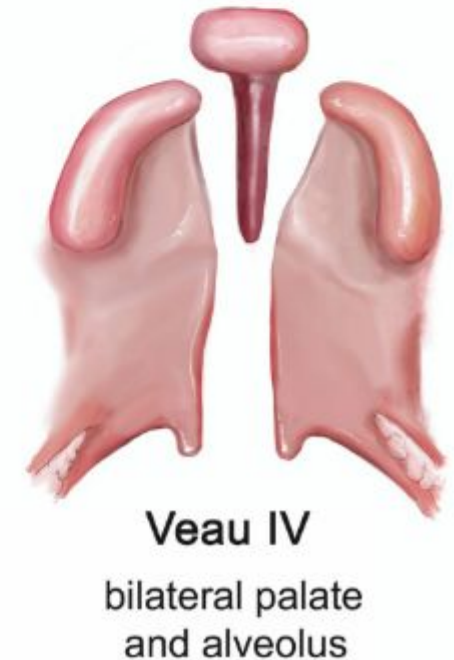
VEAU CLASS III

Complete unilateral cleft
involving lip and palate



VEAU CLASS IV

Complete bilateral cleft



ETIOLOGY

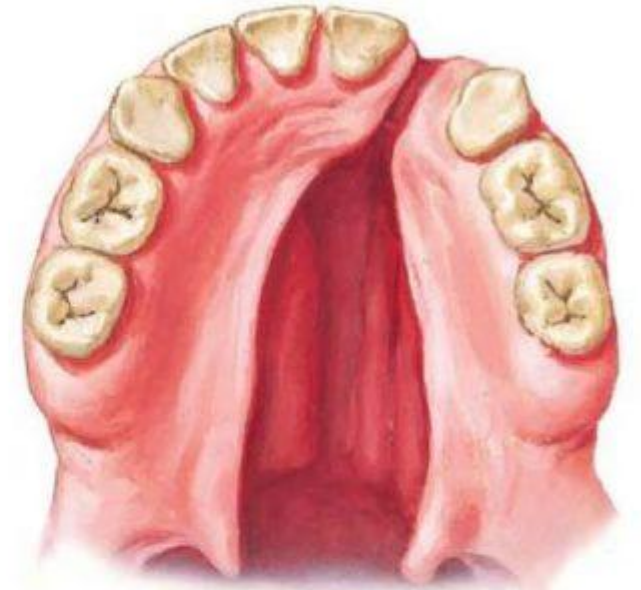
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SYNDROMIC

- Single gene mendelian
- Chromosomal defects
- Pierre-robin sequence

NON-SYNDROMIC

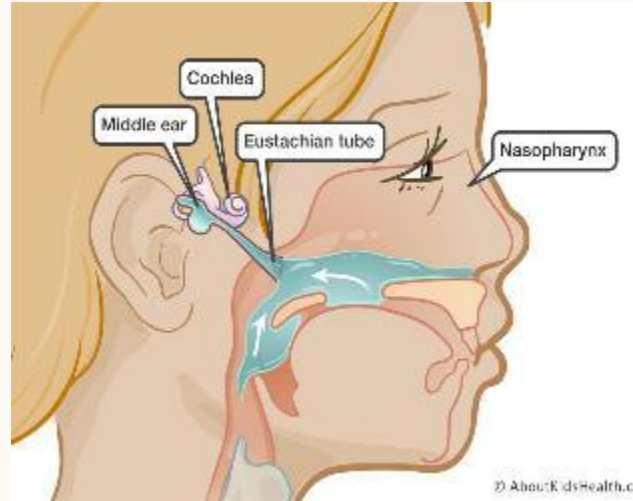
- Anticonvulsant drugs (epilepsy)
- Retinoic acid (Vit-A)
- Methotrexate - chemo drug
- Smoking, Alcohol
- Low folic acid



Complete cleft of secondary palate
and unilateral cleft of primary palate

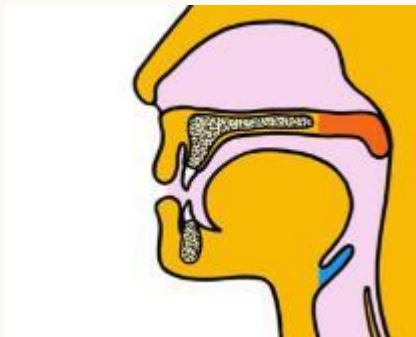
CHALLENGES AND COMPLICATIONS OF CLEFT PALATE

Speech and Language Delays



Hearing Problem

Nasal Regurgitation



when swallowed food or fluid backtracks and enters the nose

Respiratory Issues



Dental and Orthodontic Issues

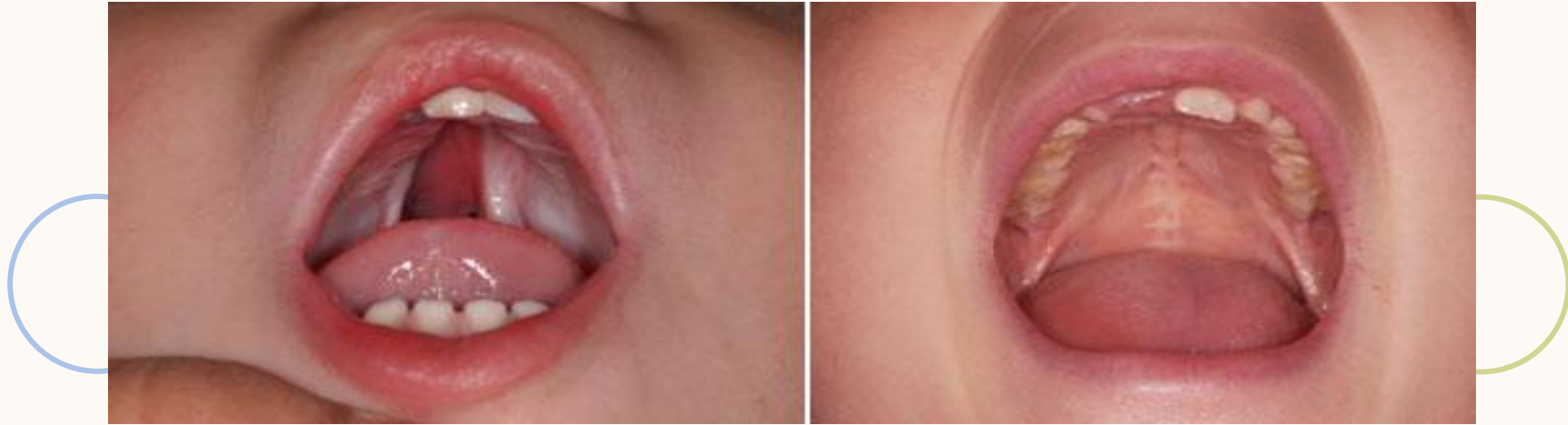


Feeding Difficulties



Feeder especially for cleft palate

TREATMENT



- Surgical repair: usually done in 6-12 months of baby
- Speech therapy
- Dental and orthodontic care
- Ear, Nose, and Throat (ENT) evaluation

RESEARCH

Prenatal Diagnosis:

Advances in medical imaging technology have allowed for it

Tissue Engineering

Regenerative medicine techniques aim to reduce the need for traditional surgical interventions in cleft palate repair, offering improved tissue integration, reduced scarring, and personalized treatment for better outcomes.

Etiology and Genetics

THANK YOU!